server2 foundation operator manual

S04 provides a Rust host foundation for local validation. It serves operational health only. Weather/geocode, S3, provider adapters, capture jobs and production cutover are later tasks. Keep nginx and production routing on legacy.

Build and check

Prerequisites: pinned Rust 1.99.0 with rustfmt/clippy, Python 3.11+, and an isolated checkout. Cargo may download public dependencies; the binary never calls providers.

```
cd server2
cargo fmt --check
cargo clippy --locked --workspace --all-targets -- -D warnings
cargo test --locked --workspace
cargo build --locked --release
python3 tools/smoke.py --binary target/release/server2
```

The smoke starts two sequential release processes on ephemeral loopback ports (PORT=0) in an empty working directory, so no real server2/.env is read. It checks GET/HEAD health, rejects public metrics, reads private counters, tests the built-in 9-second incomplete-header timeout with elapsed bounds, socket-cap (128) rejection/recovery, and both idle and partial-header SIGTERM shutdown. It also verifies that an invalid legacy PORT fails startup. No legacy collectors run.

Configure and operate

Start server2/target/release/server2 from the server2/ directory. S10e (#2753): the process reads its environment and then .env in the working directory (a copy of the legacy server/.env, see server2/.env.example) with the legacy dotenv rules; a process variable wins over the file. The listener follows legacy config.js: address OPENSHIFT_NODEJS_IP (default 127.0.0.1), port OPENSHIFT_NODEJS_PORT, else PORT, else 3000. Private metrics listen on the loopback address at the next port (an ephemeral port with port 0). Startup prints the two bound socket addresses. Invalid values and listener collisions fail startup. During coexistence on the legacy host, give server2 a port other than the legacy origin's (for example PORT=3002 in the process environment, metrics 3003).

Process limits are built in; no setting changes them (the S04 SERVER2_* overrides were removed in S10e because legacy has no equivalent and no documented operation needs another value):

Limit	Built-in value	Removed S04 variable
Worker threads	2	SERVER2_WORKER_THREADS
Blocking threads	4	SERVER2_BLOCKING_THREADS
CPU permits	2	SERVER2_CPU_LIMIT
In-flight requests	40	SERVER2_INFLIGHT_LIMIT
Volatile cache bytes	16777216	SERVER2_CACHE_BYTES
Public connections	128	SERVER2_CONNECTION_LIMIT
Connection lifetime	9 seconds; 12 with the gateway enabled	SERVER2_CONNECTION_SECONDS

Metrics connections have a separate fixed limit 16. Socket limits bound slow headers and body draining; a connection expires at its maximum age even with keep-alive. This is an absolute connection lifetime, distinct from the S10 per-request 9-second gateway deadline. When SERVER2_GATEWAY_LEGACY_URL is set, the lifetime is 12 seconds, which keeps the deadline 501 inside the connection lifetime (#2730). The guarantee covers a request that starts within lifetime minus 9 seconds of the connection being accepted; a later keep-alive request can still outlive its connection. Without the gateway the 9-second lifetime and socket behavior are unchanged. The volatile cache is a single byte-bounded slot; keyed/sharded cache algorithms are S07. CPU permits remain reserved until blocking work ends after caller cancellation.

```
curl -i http://127.0.0.1:3000/health
curl -i http://127.0.0.1:3001/internal/metrics
```

Health returns 200 with exact body OK, text/html; charset=utf-8 and CORS *. HEAD returns 200, no body, Content-Length 2. Public /internal/metrics returns 404. Private metrics returns 200 and no-store, without CORS access, with only finite counters and declared limits. OPTIONS /health currently returns 405; legacy CORS preflight (204), session-cookie and complete middleware-header parity are deferred to S02 and not asserted here; S02 freezes the complete legacy oracle before route ports. SIGTERM/SIGINT drain both listeners for at most 5 seconds; a signal-triggered timeout logs a forced stop and exits successfully. Startup, signal-registration and unexpected serving failures remain unsuccessful. Tokio runtime shutdown then waits at most 1 second for blocking work, including error paths, giving a total shutdown bound of 6 seconds plus scheduling overhead. A CPU closure may outlive that wait until process exit; this does not claim a durable S3 or provider drain. Later tasks own their interruption guarantees.

Enforce placement before task completion

Declare owned directories and exact outside exceptions in config/tasks/<task>.json before editing and link the issue declaration. Use an exact reviewed task base:

```
cd server2
python3 tools/check_placement.py --root .. --declaration config/tasks/S04.json --base <exact-task-base>
python3 tools/test_placement.py
```

The gate scans full tracked server2 files, Cargo paths, symlinks and both rename ends. Rust checks cover include*/path/cfg_attr (all delimiters and comment separators) and direct read/read_to_string/read_dir/open/create/Path/PathBuf strings. Plain/raw and borrowed/parenthesized literals are supported; encoded or unsupported compiler arguments reject. Source-relative compiler paths are verified owned separately from runtime-CWD paths. Other Rust literals retain the legacy/sibling path guard, including Command arguments. Char/byte-char escape tokens and lifetimes stay aligned; ambiguous apostrophes reject incomplete.

Python executable-call AST, decoded JSON/TOML and shell/deploy paths are checked, including executable/shebang and .bash/.zsh scripts. Only validated task outside.path identities and exact test_placement.py self.write/fixture_symlink data roles are exempt; executable calls, runtime fields and invalid declarations still reject. Root Cargo and undeclared outside implementation reject. Git/JSON/TOML failures are incomplete checks. Binary executables, computed/aliased paths, custom loaders and macro-generated runtime access need human dependency/provenance review. These literal checks are a review aid, not an adversarial sandbox. Shared workflow permits checkout@v4/full history and one owned CI command, with a 30-minute timeout. It runs only on server2-related PRs. Remote branch protection is not configured by this task.

Actual usage capture

This capture comes from the 2026-10-06 release smoke on ephemeral ports. Its manifest records platform and binary/capture hashes. This is local foundation evidence; providers, deployment and full client parity were not exercised.

```
ACTUAL RELEASE-PROCESS SMOKE | 2026-10-15
python3.11 server2/tools/smoke.py --binary server2/target/release/server2

PASS invalid legacy PORT startup rejected
server2 public=127.0.0.1:53849 metrics=127.0.0.1:53850
GET public /health -> 200 OK; text/html; CORS *
HEAD public /health -> 200; empty body; Content-Length 2
GET public /internal/metrics -> 404
GET loopback metrics -> 200 no-store; counters: {"server2_cache_bytes": "0",
"server2_cache_limit_bytes": "16777216", "server2_cpu_inflight": "0", "server2_cpu_limit": "2",
"server2_gateway_abandoned_total": "0", "server2_gateway_backend_attempts_total": "0",
"server2_gateway_legacy_geocode_attempts_total": "0", "server2_gateway_owned_inflight": "0",
"server2_gateway_requests_total": "0", "server2_gateway_responses_client_error_total": "0",
"server2_gateway_responses_dependency_error_total": "0", "server2_gateway_responses_not_modified_total":
"0", "server2_gateway_responses_ok_total": "0", "server2_gateway_responses_unavailable_total": "0",
"server2_geocoder_cache_hits_total": "0", "server2_geocoder_capacity_rejections_total": "0",
"server2_geocoder_flights_total": "0", "server2_geocoder_google_attempts_total": "0",
"server2_geocoder_joined_total": "0", "server2_geocoder_kakao_attempts_total": "0",
"server2_geocoder_quota_rejections_total": "0", "server2_health_requests_total": "2",
"server2_inflight": "0", "server2_inflight_limit": "40", "server2_rejected_total": "0"}
PASS public socket cap 128 rejects excess and recovers after release
PASS unfinished HTTP headers closed within 8.5..10.5s at the 9-second lifetime
PASS SIGTERM drained both listeners; no provider calls
PASS SIGTERM with partial headers: logged clean forced stop within 4.5..7s
```